JOHNS HOPKINS
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FPGA Design Using VHDL

Module 6C: Type Conversions

Unsigned and Signed Types

- Used to represent numeric values
  - `Unsigned`: $[0, 2^{N}-1]$ where user defines `N`
  - `Signed`: $[-2^{N-1}, 2^{N-1}-1]$, two's complement representation

- Usage similar to `std_logic_vector`:

```
signal A_unsigned : unsigned(3 downto 0);
signal B_signed   : signed (3 downto 0);
signal C_slv      : std_logic_vector (3 downto 0);
. . .
A_unsigned <= "1111" ; -- = 15 decimal
B_signed <= "1111" ;   -- = -1 decimal
C_slv <= "1111" ;      -- not decimal
```

Type Conversions (`numeric_std`) 1

- unsigned/signed ⬄ std_logic_vector
  - unsigned to std_logic_vector
    - `A_slv <= std_logic_vector(B_uv);`
  - signed to std_logic_vector
    - `C_slv <= std_logic_vector(D_sv);`
  - std_logic_vector to unsigned
    - `E_uv <= unsigned(F_slv);`
  - std_logic_vector to signed
    - `G_sv <= signed(H_slv);`
  - Note: Vectors for each assignment are same width

Type Conversions (`numeric_std`) 2

- unsigned, signed ⇔ integer
  - unsigned to integer
    - `unsigned_int <= TO_INTEGER(A_uv);`
  - signed to integer
    - `signed_int <= TO_INTEGER(B_sv);`
  - integer to unsigned
    - `C_uv <= TO_UNSIGNED(unsigned_int, 8);`
  - integer to signed
    - `D_sv <= TO_SIGNED(signed_int, 8);`

```
-- Signal definitions for above examples
signal A_uv, C_uv   : unsigned(7 downto 0);
signal unsigned_int : integer range 0 to 255;
signal B_sv, D_sv   : signed (7 downto 0);
signal signed_int   : integer range -128 to 127;
```

Type Conversions (`numeric_std`) 3

- std_logic_vector ⇔ integer (two step conversion)
  - std_logic_vector to integer
    - `unsigned_int <= to_integer(unsigned(A_slv));`
    - `signed_int <= to_integer(signed(B_slv));`
  - integer to std_logic_vector
    - `C_slv <= std_logic_vector(to_unsigned(unsigned_int, 8));`
    - `D_slv <= std_logic_vector(to_signed(signed_int, 8));`

```
signal A_slv, C_slv : std_logic_vector (7 downto 0) ;
signal Unsigned_int : integer range 0 to 255 ;
signal B_slv, D_slv : std_logic_vector( 7 downto 0) ;
signal Signed_int : integer range -128 to 127;
```